



Roll No.5019932.....

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**BCA 4 Years 3rd Semester (NEP)
Examination – December, 2025**

OPERATING SYSTEM**Paper : 24BCA403DS01****Time : Three Hours]****[Maximum Marks : 70****(50)**

Before answering the questions, candidates should ensure that they have been supplied the correct and complete question paper. No complaint in this regard, will be entertained after examination.

Note : Attempt *five* questions in all, selecting *one* question from each Unit. Question No. 1 is *compulsory*. All questions carry equal marks.

1. Answer the following in brief :

7 × 2 = 14**(13)**

- (a) Mention various functions of an operating system.
- (b) What is the cause of thrashing ? How does the system detect thrashing ?
- (c) Differentiate between preemptive and non-preemptive scheduling.
- (d) Paging vs demand paging.

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- (e) Various attributes of files.
- (f) Context switching.
- (g) Interrupt handlers.

UNIT – I

2. What do you mean by the operating system ? Explain its various types. 14 / (2)

3. Explain the following :

- (a) Types of system calls 7
- (b) Operating system structure 7

UNIT – II

4. Consider the performance of the FCFS algorithm for three compute-bound processes. What if we have processes P1 (takes 24 seconds), P2 (takes 3 seconds) and P3 (takes 3 seconds) ? If arrive in order P1, P2, P3, what is : 14 / (2)

- (i) Waiting Time ?
- (ii) Turnaround Time ?
- (iii) Throughput ?

What about if processes come in order P2, P3, P1 ?
What is :

- (i) Waiting Time ?
- (ii) Turnaround Time ?
- (iii) Throughput ?

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5. What is a Process ? Describe its various states and explain various operations performed on the processes.

14

UNIT – III

6. (a) Consider a logical address space of 256 pages with a 4-KB page size mapped onto a physical memory of 64 frames.

7

(i) How many bits are required in the logical address ?

(ii) How many bits are required in the physical address ?

(b) Consider the following page reference string : 7

7, 2, 3, 1, 2, 5, 3, 4, 6, 7, 7, 1, 0, 5, 4, 6, 2, 3, 0, 1.

Assuming demand paging with three frames, how many page faults would occur for the following replacement algorithms ?

(i) LRU replacement

(ii) FIFO replacement

(iii) Optimal replacement

7. Write notes on :

(a) Swapping

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(b) Thrashing

5

(c) Demand Paging

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(3)

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UNIT – IV

8. Discuss various directory structures with examples. 14

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9. Write notes on :

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(a) DMA

(b) Polling and Interrupt

(c) SCAN scheduling

